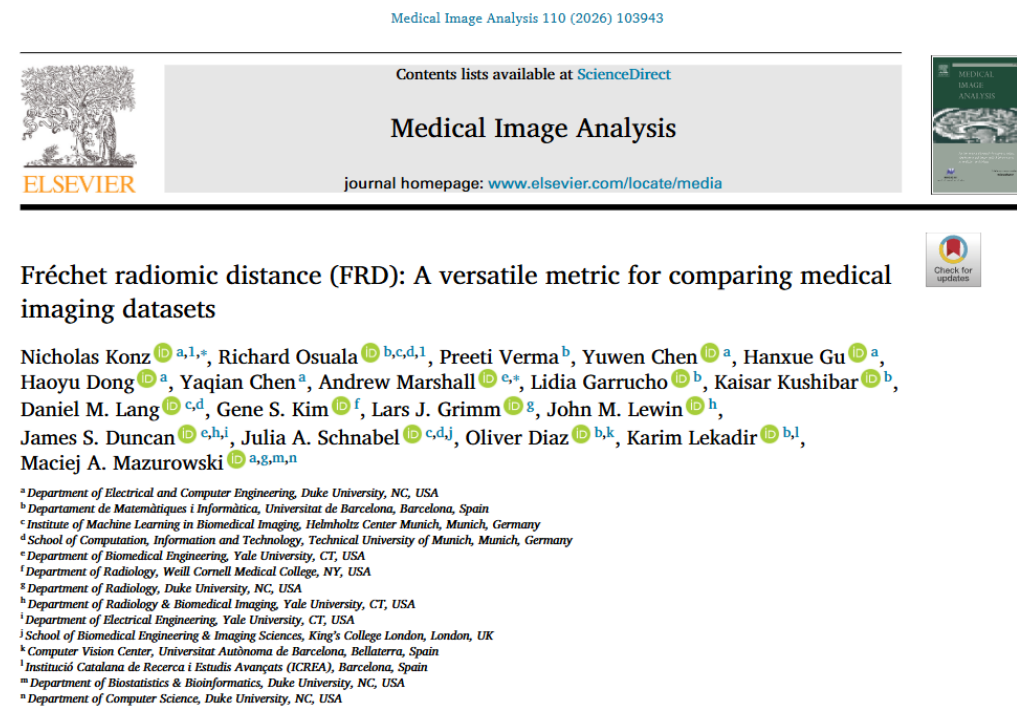


Fréchet radiomic distance (FRD): A versatile metric for comparing medicalimaging datasets

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YS Park



- What is Fréchet Inception Distance (FID)?
- Introduction
- Methods
- Evaluation and results
- Properties of FRD
- FRD for interpretability
- Discussion and conclusions

What is Fréchet Inception Distance (FID)?

- 2-Wasserstein distance = Earth Mover's Distance, EMD ([오승상 교수님 유튜브 참고](#))
 - P 분포를 Q 분포로 이동을 시키는데, 가장 효율적으로 이동하는 방법(optimal transport).

$$X_r \sim N(\mu_r, \sigma_r^2), X_g \sim N(\mu_g, \sigma_g^2)$$

$$W_2^2 = (\mu_r - \mu_g)^2 + (\sigma_r - \sigma_g)^2$$

$$\begin{aligned} (\sigma_r - \sigma_g)^2 &= \sigma_r^2 - 2\sigma_r\sigma_g + \sigma_g^2 \\ &= \Sigma_r + \Sigma_g - 2\sqrt{\Sigma_r\Sigma_g} \end{aligned}$$

$$\Sigma = \sigma^2$$

What is Fréchet Inception Distance (FID)?

- 실제 이미지 데이터셋과 생성된 이미지 데이터셋을 ImageNet의 Inception-v3 에 통과시켜 추출한 Feature들의 분포를 비교함.

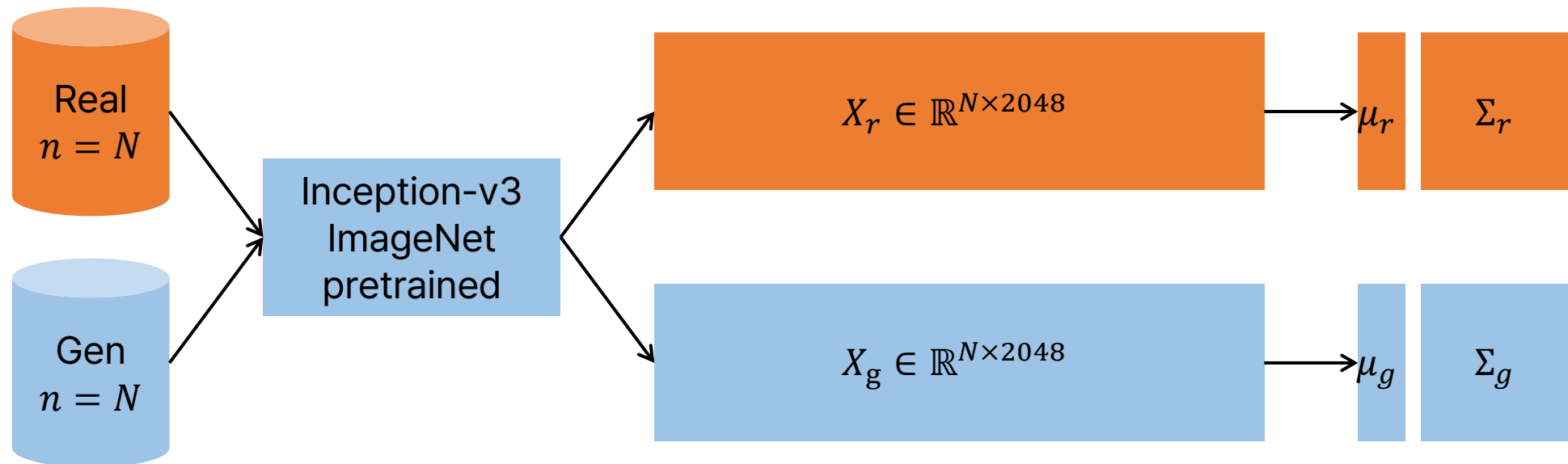
$$FID = \underbrace{|\mu_r - \mu_g|_2^2}_{\text{fidelity}} + \underbrace{\left\{ \text{Tr} \left(\Sigma_r + \Sigma_g - 2 \sqrt{\Sigma_r \Sigma_g} \right) \right\}}_{\text{diversity}}$$

- μ_r, Σ_r : 실제 이미지 feature들의 평균, 공분산
- μ_g, Σ_g : 생성 이미지 feature들의 평균, 공분산
- $|\mu_r - \mu_g|_2^2$: 두 분포 중심이 얼마나 떨어져 있는지
- $\text{Tr} \left(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{1/2} \right)$: 다양성이 실제 데이터만큼 확보됐는지

What is Fréchet Inception Distance (FID)?

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- OOD detection과 image-to-image translation 같은 DL-driven medical image analysis에서 image dist.를 비교하는 것은 중요함.
- 그러나 FID나 KID 같은 일반 컴퓨터 비전용 지표는 ImageNet 기반 특징을 사용하기 때문에 의료 영상에 그대로 적용하기엔 한계가 있음.
 - 강아지와 비행기를 비교하기 위한 feature와 정상 폐와 폐렴 폐를 비교하기 위한 feature가 같을까?
- 기존 대안인 RadFID (RadImageNet 특징 기반)도 개선은 있었지만 **해석 가능성, 소규모 데이터셋에서의 안정성** 등이 부족함.
 - Too many features 통계적 안정성 ↓
- **FRD**는 radiomics에서 널리 쓰이는 사전 정의된 해석 가능한 특징(radiomic features)을 활용하여 위 문제들 해결함.

- FID와 같은 일반적인 이미지 분포 비교 지표들이 의료 영상 고유의 요구사항을 충족시키지 못하는 한계점을 짚어냄.
- Radiomics feature에 기반한 task-independent 지각 지표인 FRD를 제안하며, 기존 지표 대비 다음과 같은 다양한 개선점을 제공함.
 - (1) downstream task와의 정합성
 - (2) 소규모 데이터셋에서의 안정성 및 계산 효율성
 - (3) 임상적 해석 가능성
 - (4) corruption 및 adversarial attack에 대한 민감도
 - (5) 방사선전문의의 이미지 품질 인식과의 일치성
- 다양한 dataset과 응용 분야에 걸쳐 FRD를 검증함.
- FRD_{v0} 을 거의 모든 테스트 시나리오에서 능가함.
- 코드 공개 및 배포 완료

- RadImageNet에서 학습된 RadFID는 FID의 대체로 사용될 수 있으나 해석가능성이 부족함.
 - Image-to-Image Translation 같은 경우 원본 이미지와 어떻게 변경되거나 다른 지에 대한 특정 질문에 답하는 것이 종종 중요함.
 - E.g. MRI→CT translation 후 "뼈 구조가 사라졌다"처럼 구체적으로 짚을 수 있어야 함.
 - 그러나 RadFID/FID에 사용된 learned deep feature는 안정적으로 해석하기 어려움.
- 해석가능한 대안인 **a space of real-valued radiomic features** of images를 활용하는 FRD 제안함.
- wavelet/frequency-space conversion 등의 영상 처리한 후 다양한 radiomics feature를 추출하여 총 464개의 feature를 추출함.
 - 참고: FID, RadFID는 2048개의 learned deep feature를 가짐.

Radiomic Feature Extraction

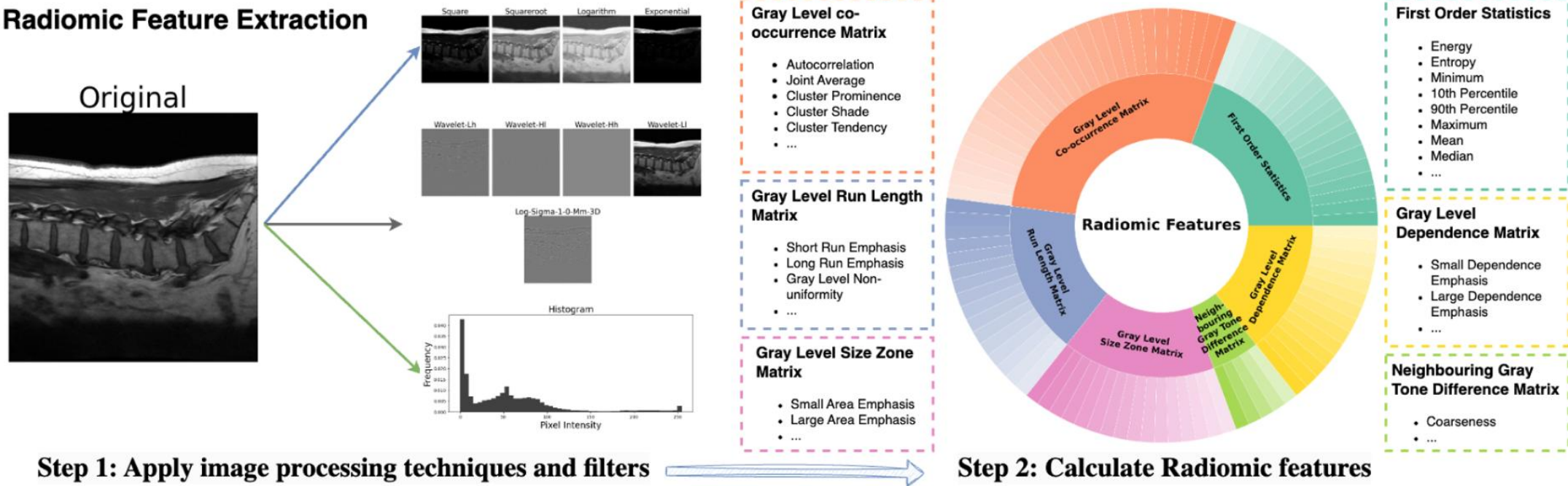


Fig. 3. Our extraction process and taxonomy for radiomic image features. From left to right, the steps of feature extraction include (1) the passing of the input image through possible filters, *e.g.*, wavelet/frequency-space conversion, and (2) the measurement of various first-order or higher-order features, *e.g.*, entropy and gray-level matrix features, respectively.

Filter Classes

Aside from the feature classes, there are also some built-in optional filters:

- Laplacian of Gaussian (LoG, based on SimpleITK functionality)
- Wavelet (using the PyWavelets package)
- Square
- Square Root
- Logarithm
- Exponential
- Gradient (Magnitude)
- Local Binary Pattern (LBP) 2D / 3D

Feature Classes

Currently supports the following feature classes:

- First Order Statistics
- Shape-based (2D and 3D)
- Gray Level Co-occurrence Matrix (GLCM)
- Gray Level Run Length Matrix (GLRLM)
- Gray Level Size Zone Matrix (GLSZM)
- Gray Level Dependence Matrix (GLDM)
- Neighboring Gray Tone Difference Matrix (NGTDM)

- **Imaging Biomarker Standardization Initiative(IBSI)** 정의를 준수한 radiomics feature
 - **First Order Statistics** (~19개):
 - 개별 픽셀의 intensity 값 분포(histogram)만으로 계산하는 가장 기본적인 통계량
 - **Gray Level Co-occurrence Matrix (GLCM, ~24개)**
 - 특정 방향과 거리에 있는 픽셀 쌍의 intensity 조합이 얼마나 자주 나타나는지를 세는 행렬
 - **Gray Level Run Length Matrix (GLRLM, ~16개)**
 - 특정 방향으로 동일한 intensity 값이 연속(run)해서 몇 번 나타나는지를 세는 행렬
 - **Gray Level Size Zone Matrix (GLSZM, ~16개)**
 - 방향에 무관하게 연결된 동일 intensity 영역의 크기를 세는 행렬
 - **Gray Level Dependence Matrix (GLDM, ~14개)**
 - 중심 픽셀과 neighborhood 중 intensity 차이가 α 이내인 픽셀 개수를 세는 행렬
 - **Neighbouring Gray Tone Difference Matrix (NGTDM, ~5개)**
 - 각 픽셀의 intensity와 neighborhood 픽셀들의 평균 intensity 간 차이의 절댓값 합을 gray level별로 누적한 행렬

$$FRD(D_1, D_2) = \log \sqrt{|\mu_{D_1} - \mu_{D_2}|_2^2 + \text{Tr} \left(\Sigma_{D_1} + \Sigma_{D_2} - 2\sqrt{\Sigma_{D_1}\Sigma_{D_2}} \right)}$$

- 각 feature를 D_1 에 대하여 z-score normalize한 후 계산
- MMD distance도 시도해봤으나 Fréchet에 비해 미미

- FRD는 downstream task와 무관하게 분포 자체만 비교함.
- 의료 영상에서는 task-dependent 지표가 더 일반적임.
 - 평가 방식 (2가지 방향)
 - 실제 타겟 도메인으로 학습 → 변환 이미지로 테스트
 - 반대로 변환 이미지로 학습 → 실제 이미지로 테스트
 - segmentation 인 경우 anatomical consistency도 함께 측정 가능함.

작업	지표
Segmentation	Dice coefficient
Object detection	mIoU, mAP@[0.5, 0.95]
Classification	AUC

Table 1

Main datasets evaluated in this paper. “Domains” are the source→target domain pairs used, e.g., for image translation. Each dataset is labeled for how its domains are similar (“Intra-”) and how they differ (“Inter-”), e.g., the BraTS dataset has intra-modality, inter-sequence domains. “FGT” is fibroglandular tissue.

Abbrev. name	Full name/citation	Domains	Intra-	Inter-	Train/val/test sizes	Downstream tasks
Breast MRI	Duke Breast CancerMRI (DBC)(Saha et al., 2018a),(Lew et al., 2024)	Siemens→GE(T1 MRI)	Sequence	ScannerManuf.	12K/2.4K/2.6K	FGT segmentation,breast segmentation,cancer classification
Brain MRI	BraTS(Menze et al., 2015)	T1→T2	Modality	Sequence	28K/6K/6K	Tumor segmentation,tumor detection,cancer classification
Lumbar spine	TotalSegmentator(Wasserthal et al., 2023)and in-house MRIs	T1 MRI→CT	Body region	Modality	2K/0.6K/0.6K	Bone segmentation
CHAOS	CHAOS(Abdom. MRI & CT)(Kavur et al., 2021)	CT→T1 MRI(in-phase)	Body region	Modality	1.8K/1.1K/0.6K	Liver segmentation,liver classification

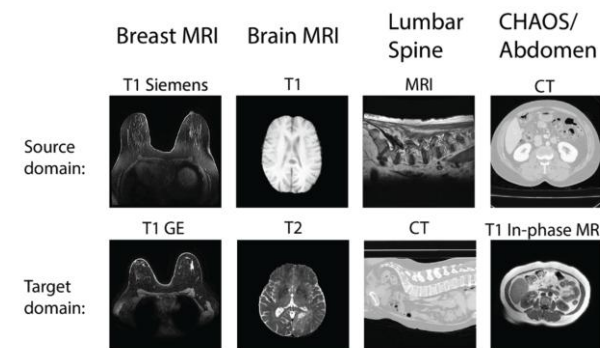


Fig. 4. Example images from each dataset, ordered left-to-right with respect to Table 1.

Table 3

Perceptual/task-independent metrics $d(D_{s \rightarrow t}^{\text{test}}, D_t^{\text{test}})$ for image translation models. Best and runner-up models according to each metric in bold and underlined, respectively.

Breast MRI							Brain MRI					
Method	FRD	FRD _{v0}	RadFID	FID	KID	CMMD	FRD	FRD _{v0}	RadFID	FID	KID	CMMD
CycleGAN	38.1	706	0.26	107	0.049	0.308	33.4	554	0.06	21.7	<u>0.004</u>	0.378
MUNIT	43.7	626	0.29	144	0.089	1.480	<u>25.7</u>	540	<u>0.05</u>	<u>21.6</u>	0.006	0.388
CUT	<u>24.8</u>	<u>632</u>	0.17	106	0.053	0.362	33.8	552	0.13	29.4	0.012	<u>0.259</u>
GcGAN	24.6	647	0.17	<u>104</u>	<u>0.040</u>	<u>0.322</u>	12.1	523	0.04	19.0	0.003	0.239
MaskGAN	48.7	1042	0.35	118	0.089	0.642	27.8	555	0.06	23.5	0.008	0.392
UNSB	24.6	645	<u>0.19</u>	91	0.033	0.388	12.1	<u>525</u>	0.08	26.0	0.010	0.563

Lumbar						CHAOS						
Method	FRD	FRD _{v0}	RadFID	FID	KID	CMMD	FRD	FRD _{v0}	RadFID	FID	KID	CMMD
CycleGAN	6.71	<u>350</u>	0.25	210	0.161	2.950	42.8	470	<u>0.11</u>	122	0.051	<u>0.379</u>
MUNIT	9.31	367	0.30	197	0.151	2.317	5.41	276	0.10	136	0.073	0.904
CUT	6.48	417	0.21	245	0.206	3.373	6.84	514	0.10	145	0.083	0.444
GcGAN	<u>6.52</u>	313	0.25	226	<u>0.161</u>	3.300	<u>6.38</u>	<u>434</u>	0.12	141	<u>0.064</u>	0.507
MaskGAN	6.64	421	0.27	248	0.217	3.237	58.8	437	0.22	212	0.130	2.120
UNSB	6.59	375	<u>0.23</u>	<u>208</u>	0.172	<u>2.579</u>	51.8	542	<u>0.11</u>	<u>135</u>	0.078	0.356

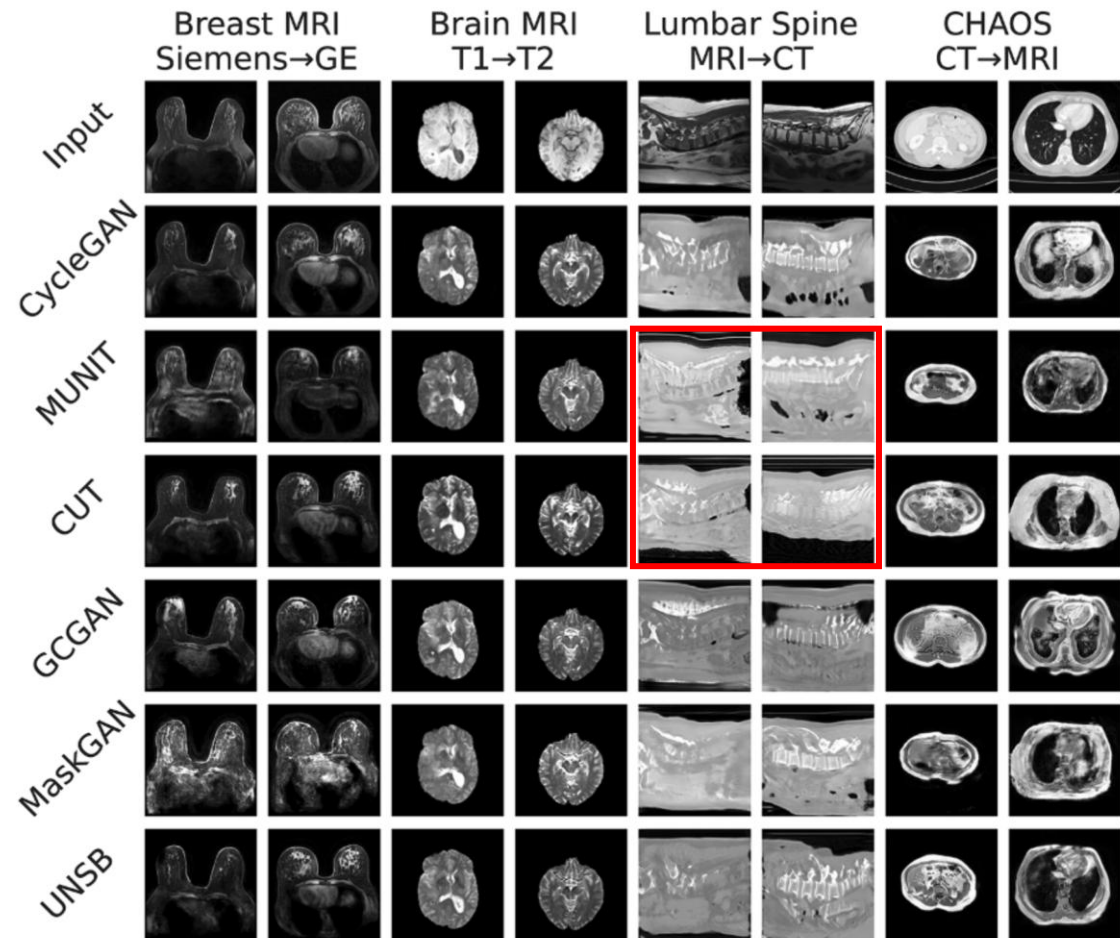


Fig. 6. Translations $x_{s \rightarrow t}^{\text{test}}$ from each translation model (non-top rows) given example inputs x_s^{test} (top row).

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Table 4

Downstream task performance metrics $\text{Perf}(D_{s \rightarrow t}^{\text{test}})$ for image translation models. In-domain and out-of-domain performance shown at the bottom for reference for how susceptible each task is to domain shift, and as expected upper and lower performance bounds.

Breast MRI				Brain MRI				Lumbar	CHAOS	
Dice		AUC		Dice		mIoU		AUC	Dice	AUC
Method	Breast	FGT	Cancer	Tumor	Tumor	mAP	Cancer	Bone	Liver	Liver
CycleGAN	0.871	0.494	0.530	0.348	0.164	0.126	0.805	0.232	0.284	0.591
MUNIT	0.832	0.201	0.511	0.337	0.168	0.125	<u>0.844</u>	0.101	0.182	0.323
CUT	0.843	0.373	<u>0.544</u>	0.303	0.159	0.133	0.861	0.277	0.444	0.744
GcGAN	<u>0.876</u>	<u>0.389</u>	0.492	<u>0.360</u>	0.165	0.137	0.835	0.126	0.167	<u>0.702</u>
MaskGAN	0.809	0.164	0.441	0.375	0.170	0.126	0.842	0.167	0.317	0.375
UNSB	0.881	0.308	0.594	0.353	<u>0.169</u>	<u>0.135</u>	0.839	0.138	<u>0.381</u>	0.405

<i>In-domain</i>	0.883	0.696	0.670	0.442	0.174	0.169	0.841	0.949	0.864	0.866
<i>OOD</i>	0.747	0.446	0.538	0.005	0.152	0.065	0.727	0.007	0.062	0.504

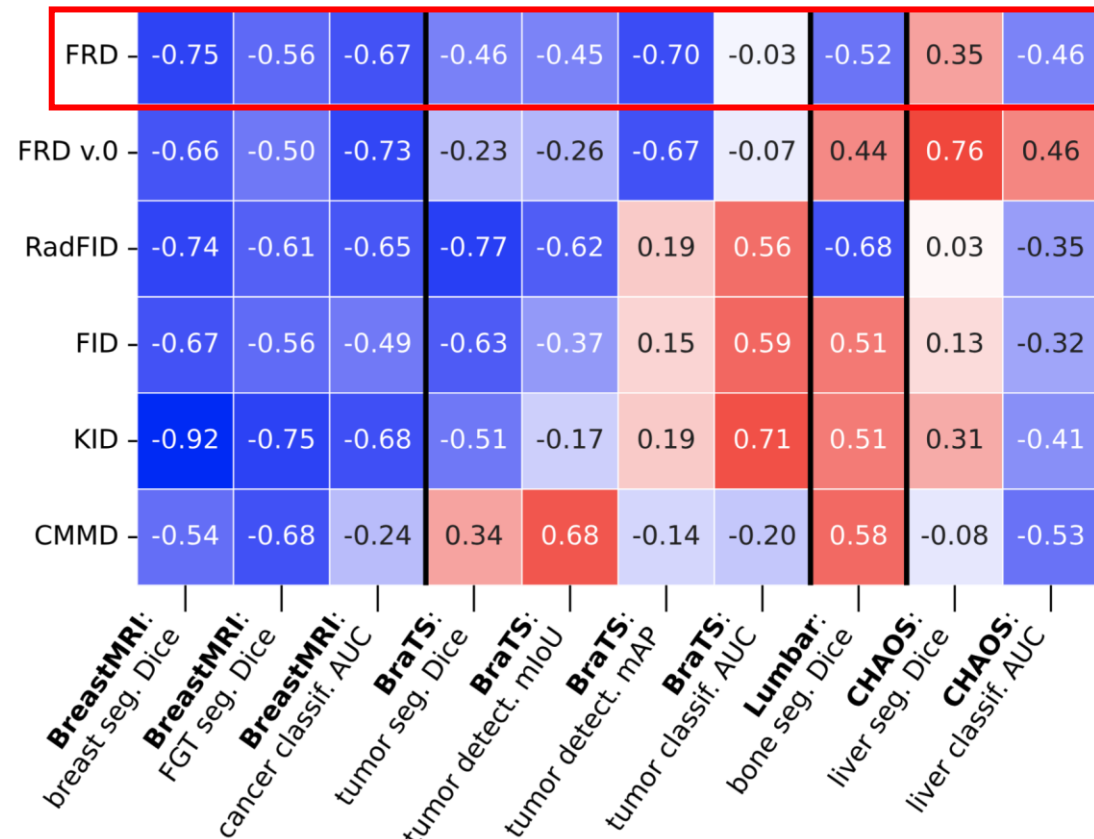


Fig. 7. Pearson correlation of perceptual metrics (vertical axis) (Table 3) with downstream task-based metrics (horizontal axis) (Table 4) for evaluating image translation, taken across all translation models (lower r (colder color) is better).

	Breast MRI		Brain MRI		Lumbar CT		Abdom. CT	
Model:	A	B	A	B	A	B	A	B
FID	67	54	157	26	210	103	232	118
KID	0.02	0.01	0.14	0.01	0.18	0.04	0.23	0.08
CMMD	0.30	0.33	0.82	0.28	3.57	0.87	4.72	0.48
RadFID	0.16	0.06	0.16	0.04	0.20	0.11	0.40	0.16
FRD _{v0}	585	568	709	633	232	167	897	710
FRD	47.4	43.4	58.5	56.3	8.3	6.3	71.3	68.6

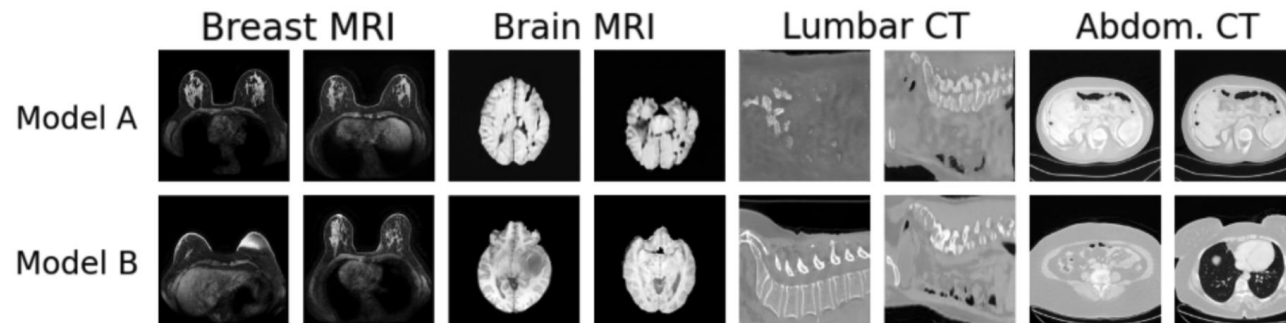


Fig. 8. Top: FRD and other perceptual metrics for evaluating unconditional generative models, comparing a poor model (A) to a better model (B). **Bottom:** example generated images.

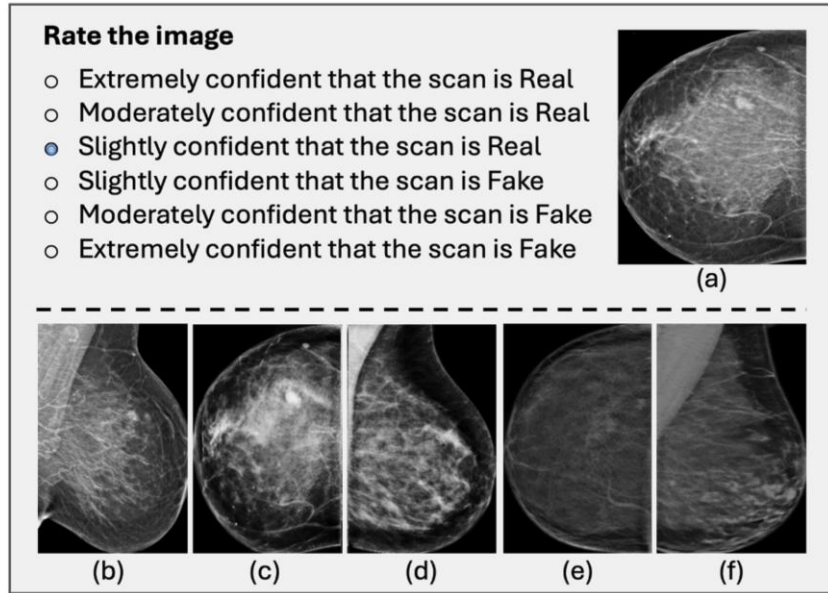


Fig. 9. Reader Study: Using an ImageJ plugin, each reader was shown one image at a time, which was randomly sampled from a set of 90 synthetic mammograms. The image set was equally distributed between craniocaudal (CC) and mediolateral oblique (MLO) views (45 each), as well as between the Optimam, CSAW, and BCDR datasets (30 each). Images (a) to (f) are synthetic samples that looked realistic to the readers: (a) BCDR CC, (b) BCDR MLO, (c) Optimam CC, (d) Optimam MLO, (e) CSAW CC, and (f) CSAW MLO. Figure based on [Garrucho et al. \(2023\)](#).

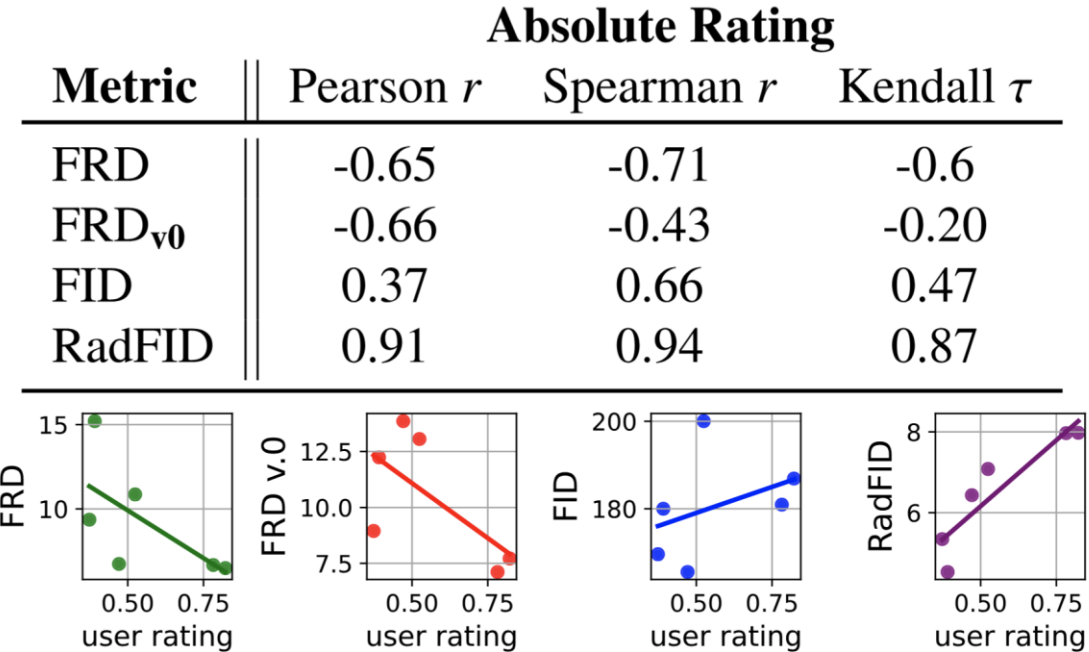


Fig. 10. Top: Correlation coefficients (linear Pearson r_p and non-linear/rank Spearman r_s and Kendall τ) of different distance metrics with average user (radiologist) preference, for the task of measuring synthetic image quality. **Bottom:** Associated plots and linear best fits for this data.

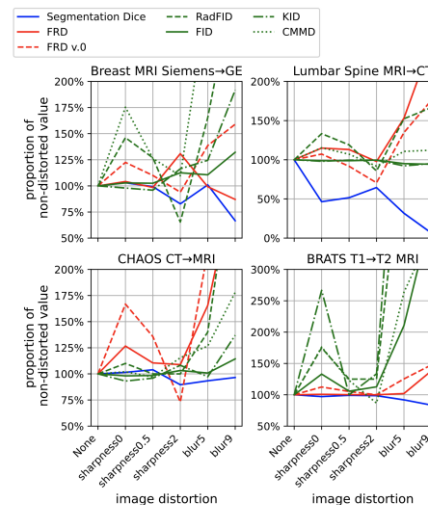
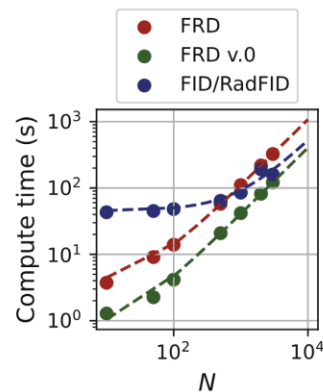
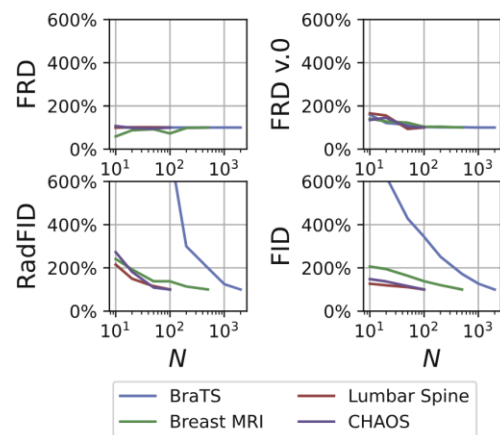


Fig. 11. Left: Sensitivity of FRD, FRD_{v0} , RadFID and FID to sample size N . Metric values (vert. axes) are relative to their highest- N result. Right: Computation time for the metrics w.r.t N , with linear best fits plotted in dashed lines.

Fig. 12. Sensitivity of FRD, FRD_{v0} (red lines), RadFID, FID, KID, and CMMD (green lines) to corruptions which affect downstream task performance (blue line) on translated images. Metric values (vertical axis) are relative to their undistorted result ("None").

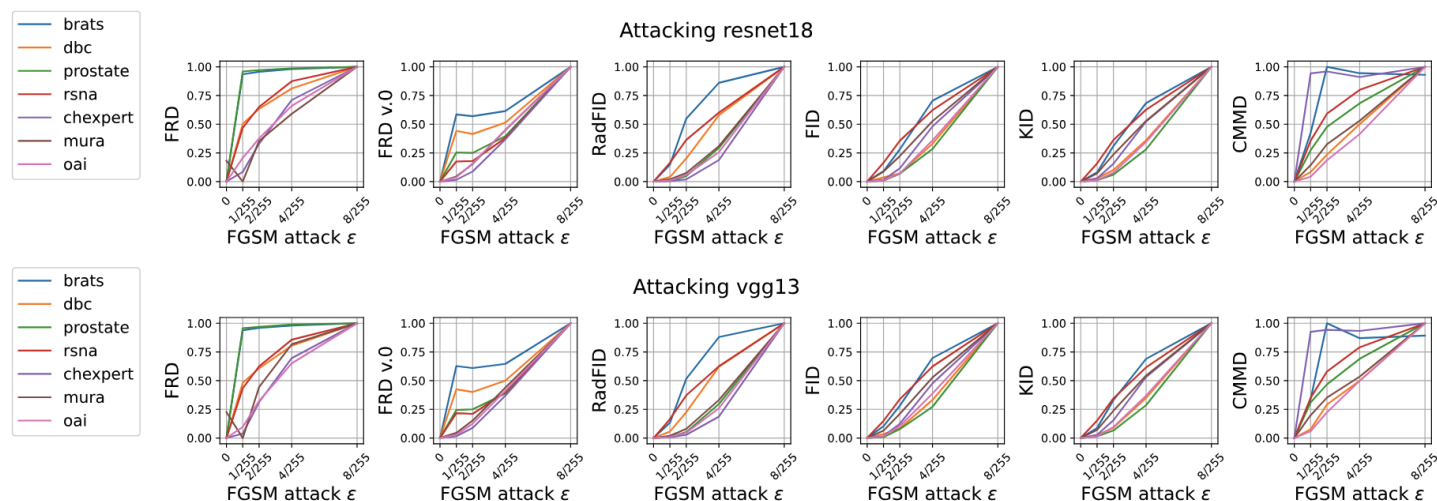


Fig. 13. Using different distance metrics (vertical axes) to detect FGSM adversarial attacks of various strength (horizontal axes), for ResNet-18 (top) and VGG-13 (bottom) classification models. Note that distance metrics are scaled to $[0, 1]$ over their plotted range of values for the sake of visualization.

- 두 영상 분포의 차이를 feature 변화 벡터 Δh 로 요약 가능함.

$$\Delta h = h_{s \rightarrow t} - h_s$$

$$\Delta h = \mu_{s \rightarrow t} - \mu_s$$

- Δh 는 feature space 상의 선형 방향 벡터로, linear representation hypothesis 계열의 해석 방법과 같은 틀임.

- The Linear Representation Hypothesis and the Geometry of Large Language Models (Park et al.)
- TCAV (Kim et al.), linear probe (Alain & Bengio)

- Radiomic feature는 intensity/shape/texture처럼 정의가 명확하기에 Δh 를 성분별로 그대로 읽어 정량적 질문에 답할 수 있음.

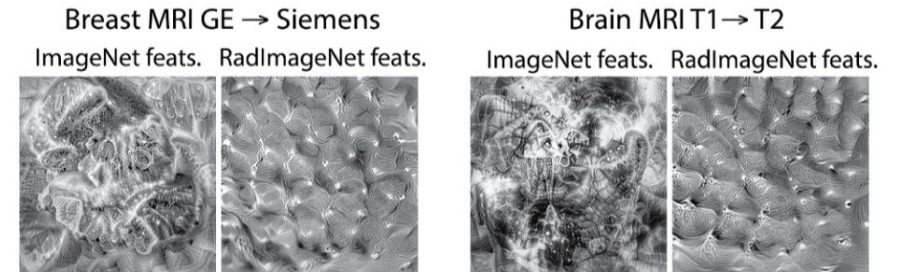


Fig. 14. Attempts at medical image translation interpretability via learned feature inversion.

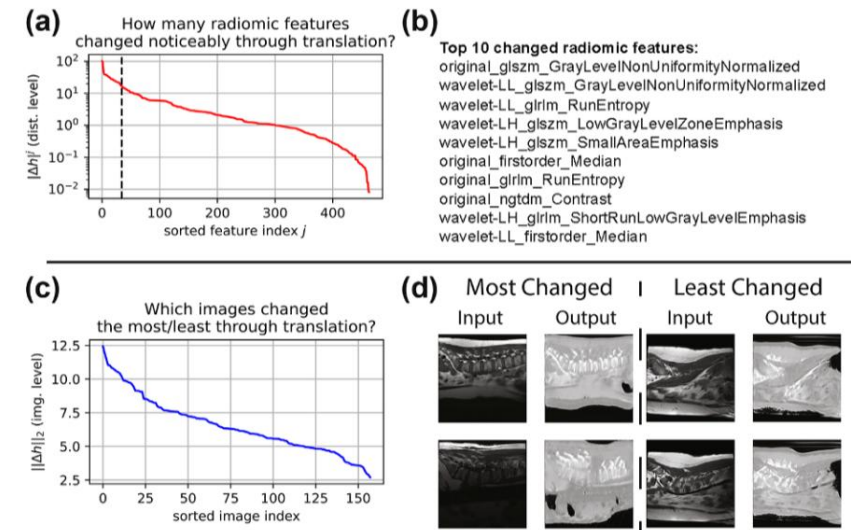


Fig. 15. Translation interpretability using radiomic features.

- Radiomics feature에 기반한 task-independent 지각 지표인 FRD를 제안하며, 기존 지표 대비 다음과 같은 다양한 개선점을 제공함.
 - (1) downstream task와의 정합성
 - (2) 소규모 데이터셋에서의 안정성 및 계산 효율성
 - (3) 임상적 해석 가능성
 - (4) corruption 및 adversarial attack에 대한 민감도
 - (5) 방사선전문의의 이미지 품질 인식과의 일치성
- Limitations
 - **radiology**에 한정됨. 다른 modality엔 radiomic feature를 조정하거나 다른 biomarker로 대체해야 할 수 있음.
 - histopathology의 **pathomics feature**로 Fréchet Pathomic Distance
 - dermatology 등에도 radiomic이 이미 유효하니 직접 적용 가능.
 - interpretability 기여는 아직 초기 단계이고 더 구체적 통찰을 뽑는 후속 연구가 필요함.
 - FID처럼 **FRD의 절댓값은 같은 데이터셋 내 상대 비교에서만** 의미가 있음.